Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$\omega_{\alpha\beta\chi}$	$\omega_{\beta\alpha\chi} == - \omega_{\alpha\beta\chi}$	$\omega_{0^-}^{\#1}{}_{\alpha\beta\chi} - \frac{1}{2} \eta_{\beta\chi} \omega_{1^-}^{\#1}{}_{\alpha} + \frac{1}{2} \eta_{\alpha\chi} \omega_{1^-}^{\#1}{}_{\beta} + \frac{4}{3} \omega_{2^-}^{\#1}{}_{\alpha\beta\chi} + \frac{1}{3} \eta_{\beta\chi} \omega_{0^+}^{\#1} n_{\alpha} + \omega_{1^+}^{\#1}{}_{\beta\chi} n_{\alpha} + \omega_{2^+}^{\#1}{}_{\beta\chi} n_{\alpha} - \frac{1}{3} \eta_{\alpha\chi} \omega_{0^+}^{\#1} n_{\beta} - \omega_{1^+}^{\#1}{}_{\alpha\chi} n_{\beta} - \omega_{2^+}^{\#1}{}_{\alpha\chi} n_{\beta} + \omega_{1^+}^{\#2}{}_{\alpha\beta} n_{\chi} - \frac{1}{2} \omega_{1^-}^{\#1}{}_{\beta} n_{\alpha} n_{\chi} - \omega_{1^-}^{\#2}{}_{\beta} n_{\alpha} n_{\chi} + \frac{1}{2} \omega_{1^-}^{\#1}{}_{\alpha} n_{\beta} n_{\chi} + \omega_{1^-}^{\#2}{}_{\alpha} n_{\beta} n_{\chi}$	$\sigma_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$\omega_{0^+}^{\#1}$	(None)	$\omega_{\alpha}{}^{\beta}{}_{\beta} n^{\alpha}$	$\sigma_{0^+}^{\#1}$
$\omega_{0^-}^{\#1}{}_{\alpha\beta\chi}$	$\omega_{0^-}^{\#1}{}_{\alpha\chi\beta} == - \omega_{0^-}^{\#1}{}_{\alpha\beta\chi}$ && $\omega_{0^-}^{\#1}{}_{\beta\alpha\chi} == - \omega_{0^-}^{\#1}{}_{\alpha\beta\chi}$	$\frac{1}{3} \omega_{\alpha\beta\chi} - \frac{1}{3} \omega_{\alpha\chi\beta} + \frac{1}{3} \omega_{\beta\chi\alpha} - \frac{1}{3} \omega_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{3} \omega_{\beta\delta\chi} n_{\alpha} n^{\delta} - \frac{1}{3} \omega_{\chi\delta\beta} n_{\alpha} n^{\delta} + \frac{1}{3} \omega_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{3} \omega_{\alpha\delta\chi} n_{\beta} n^{\delta} + \frac{1}{3} \omega_{\chi\delta\alpha} n_{\beta} n^{\delta} - \frac{1}{3} \omega_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{3} \omega_{\alpha\delta\beta} n_{\chi} n^{\delta} - \frac{1}{3} \omega_{\beta\delta\alpha} n_{\chi} n^{\delta}$	$\sigma_{0^-}^{\#1}{}_{\alpha\beta\chi}$
$\omega_{1^+}^{\#1}{}_{\alpha\beta}$	$\omega_{1^+}^{\#1}{}_{\beta\alpha} == - \omega_{1^+}^{\#1}{}_{\alpha\beta}$	$-\frac{1}{2} \omega_{\alpha\chi\beta} n^{\chi} + \frac{1}{2} \omega_{\beta\chi\alpha} n^{\chi} - \frac{1}{2} \omega_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \omega_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta}$	$\sigma_{1^+}^{\#1}{}_{\alpha\beta}$
$\omega_{1^+}^{\#2}{}_{\alpha\beta}$	$\omega_{1^+}^{\#2}{}_{\beta\alpha} == - \omega_{1^+}^{\#2}{}_{\alpha\beta}$	$\omega_{\alpha\beta\chi} n^{\chi} + \omega_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} - \omega_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta}$	$\sigma_{1^+}^{\#2}{}_{\alpha\beta}$
$\omega_{1^-}^{\#1}{}_{\alpha}$	(None)	$- \omega_{\alpha}{}^{\beta}{}_{\beta} + \omega_{\beta}{}^{\chi}{}_{\chi} n_{\alpha} n^{\beta} + \omega_{\alpha\beta\chi} n^{\beta} n^{\chi}$	$\sigma_{1^-}^{\#1}{}_{\alpha}$
$\omega_{1^-}^{\#2}{}_{\alpha}$	(None)	$\omega_{\alpha\beta\chi} n^{\beta} n^{\chi}$	$\sigma_{1^-}^{\#2}{}_{\alpha}$
$\omega_{2^+}^{\#1}{}_{\alpha\beta}$	$\omega_{2^+}^{\#1}{}_{\beta\alpha} == \omega_{2^+}^{\#1}{}_{\alpha\beta}$	$-\frac{1}{2} \omega_{\alpha\chi\beta} n^{\chi} - \frac{1}{2} \omega_{\beta\chi\alpha} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} \omega_{\chi}{}^{\delta}{}_{\delta} n^{\chi} + \frac{1}{3} \omega_{\chi}{}^{\delta}{}_{\delta} n_{\alpha} n_{\beta} n^{\chi} + \frac{1}{2} \omega_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \omega_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta}$	$\sigma_{2^+}^{\#1}{}_{\alpha\beta}$
$\omega_{2^-}^{\#1}{}_{\alpha\beta\chi}$	$\omega_{2^-}^{\#1}{}_{\beta\alpha\chi} == - \omega_{2^-}^{\#1}{}_{\alpha\beta\chi}$	$\frac{1}{2} \omega_{\alpha\beta\chi} + \frac{1}{4} \omega_{\alpha\chi\beta} - \frac{3}{8} \eta_{\beta\chi} \omega_{\alpha}{}^{\delta}{}_{\delta} - \frac{1}{4} \omega_{\beta\chi\alpha} + \frac{3}{8} \eta_{\alpha\chi} \omega_{\beta}{}^{\delta}{}_{\delta} - \frac{3}{8} \omega_{\beta}{}^{\delta}{}_{\delta} n_{\alpha} n_{\chi} + \frac{3}{8} \omega_{\alpha}{}^{\delta}{}_{\delta} n_{\beta} n_{\chi} + \frac{1}{4} \omega_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{2} \omega_{\beta\delta\chi} n_{\alpha} n^{\delta} + \frac{1}{4} \omega_{\chi\delta\beta} n_{\alpha} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \omega_{\delta}{}^{\epsilon}{}_{\epsilon} n_{\alpha} n^{\delta} - \frac{1}{4} \omega_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{2} \omega_{\alpha\delta\chi} n_{\beta} n^{\delta} - \frac{1}{4} \omega_{\chi\delta\alpha} n_{\beta} n^{\delta} - \frac{3}{8} \eta_{\alpha\chi} \omega_{\delta}{}^{\epsilon}{}_{\epsilon} n_{\beta} n^{\delta} - \frac{1}{2} \omega_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{4} \omega_{\alpha\delta\beta} n_{\chi} n^{\delta} + \frac{1}{4} \omega_{\beta\delta\alpha} n_{\chi} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \omega_{\alpha\delta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \eta_{\alpha\chi} \omega_{\beta\delta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \omega_{\beta\delta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{3}{8} \omega_{\alpha\delta\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon}$	$\sigma_{2^-}^{\#1}{}_{\alpha\beta\chi}$